

Project Initialization and Planning Phase

Date	02 July 2026
Team ID	XXXXXXX
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to assess and predict a country's human development status using machine learning, boosting the efficiency of socio-economic research and policy planning. It tackles the inefficiencies of manual HDI calculations, promising faster insights, accurate tier classifications, and better decision-making tools. Key features include a machine learning-based HDI classification model and a real-time interactive web interface for prediction.

Project Overview

Field	Details
Objective	The primary objective is to develop a predictive machine learning model that accurately classifies countries into four HDI tiers (Very High, High, Medium, Low) based on life expectancy, education, and income indicators.
Scope	The project comprehensively assesses global development data, incorporates machine learning for HDI tier prediction, and provides a web-based interface for policymakers and researchers to simulate and evaluate human development outcomes.

Problem Statement

Field	Details
Description	Addressing the lack of accessible, real-time predictive tools for evaluating multi-dimensional socio-economic data (health, education, income) which currently requires complex manual calculations and static report analysis.
Impact	Solving these issues will result in improved efficiency for policymakers and researchers, enabling data-driven decision-making, better resource allocation, and a deeper understanding of global development gaps.

Proposed Solution

Field	Details
Approach	Employing supervised machine learning classification techniques to analyze historical UNDP/World Bank data and predict a country's HDI tier based on input indicators.
Key Features	• Implementation of a machine learning-based HDI tier classification model. • Real-time decision-making via an interactive Flask web application. • Interactive data visualizations to compare user inputs against global development benchmarks.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications	Intel Core i3 or above
Memory	RAM specifications	Minimum 4 GB (8 GB recommended)
Storage	Disk space for data, models, and logs	Minimum 10 GB free space
Software		
Frameworks	Python web frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm, Anaconda Navigator
Data	Source, size, format	UNDP Human Development Reports / World Bank Open Data, CSV format, historical data for 190+ countries across multiple